# Ideation Phase
# Brainstorm & Idea Prioritization Template

| | |
|---|---|
| Date | 31 October 2026 |
| Team ID | Individual |
| Project Name | Electric Vehicle Market Performance and Insights |
| Maximum Marks | 4 Marks |

**Brainstorm & Idea Prioritization Template:**

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

## Step 1: Brainstorm and Idea Listing

Each team member lists out as many ideas as possible without judging them at this stage.

| S.No | Team Member | Idea / Suggestion | Category | Group No. |
|---|---|---|---|---|
| 1 | Vishwas Choyal | Nine-view interactive Tableau dashboard | BI dashboard | 1 |
| 2 | Vishwas Choyal | Spreadsheet with static summaries | Spreadsheet | 2 |
| 3 | Vishwas Choyal | Battery comparison report | Research report | 3 |
| 4 | Vishwas Choyal | Machine-learning classifier | Predictive modelling | 4 |

## Step 2: Idea Prioritization

Rate each grouped idea on feasibility and importance, then select the final idea(s) to move forward with.

| Group No. | Final Idea | Feasibility (High/ Medium/Lo w) | Importance (High/ Medium/Lo w) | Priority | Selected (Yes/No) |
|---|---|---|---|---|---|
| 1 | Interactive Tableau dashboard | High | High | 1 | Yes |
| 2 | Static spreadsheet report | High | Medium | 2 | No |
| 3 | Predictive classifier | Medium | Low | 3 | No |

## Selection rationale

The dashboard directly addresses the three supplied scenarios and the nine-visualization deliverable. Classification would answer a different question and require a separate holdout evaluation.

## Project author

Vishwas Choyal | EN25CS3011066 | Individual project

# Ideation Phase
# Define the Problem Statements

| | |
|---|---|
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| Maximum Marks | 2 Marks |

**Customer Problem Statement Template:**

Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love.

A well-articulated customer problem statement allows you and your team to find the ideal solution for the challenges your customers face. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

| | | |
|---|---|---|
| **I am** | Describe customer with 3-4 key characteristics - *who are they?* | Describe the customer and their attributes here |
| **I'm trying to** | *List their outcome or "job" the care about - what are they trying to achieve?* | List the thing they are trying to achieve here |
| **but** | Describe what problems or barriers stand in the way – *what bothers them most?* | Describe the problems or barriers that get in the way here |
| **because** | Enter the "root cause" of why the problem or barrier exists – *what needs to be solved?* | Describe the reason the problems or barriers exist |
| **which makes me feel** | Describe the emotions from the customer's point of view – *how does it impact them emotionally?* | Describe the emotions the result from experiencing the problems or barriers |

Reference: https://miro.com/templates/customer-problem-statement/

**Example:**



| Problem Statement (PS) | I am (Customer) | I'm trying to | But | Because | Which makes me feel |
|---|---|---|---|---|---|
| PS-1 | Emily Johnson, market strategy director | Prioritize EV product segments | Sales, ratings and range are scattered | There is no shared view of segment performance | Uncertain about portfolio priorities |
| PS-2 | David Chen, BI manager | Compare manufacturers and production geography | Totals and shares are hard to compare | Raw CSVs lack an interactive analytical view | Concerned about inconsistent client advice |
| PS-3 | Dr Sarah Williams, researcher | Compare battery technology and mobility outcomes | Aggregate trends can conceal category differences | Technical metrics and adoption data are separated | Cautious about overinterpreting evidence |

## Evidence boundary

These personas come from the supplied project scenarios. They are not interview participants. This project analyzes a synthetic dataset and does not claim real-world market outcomes.

## Project author

Vishwas Choyal | EN25CS3011066 | Individual project

# Ideation Phase
# Empathize & Discover

| | |
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| Maximum Marks | 4 Marks |

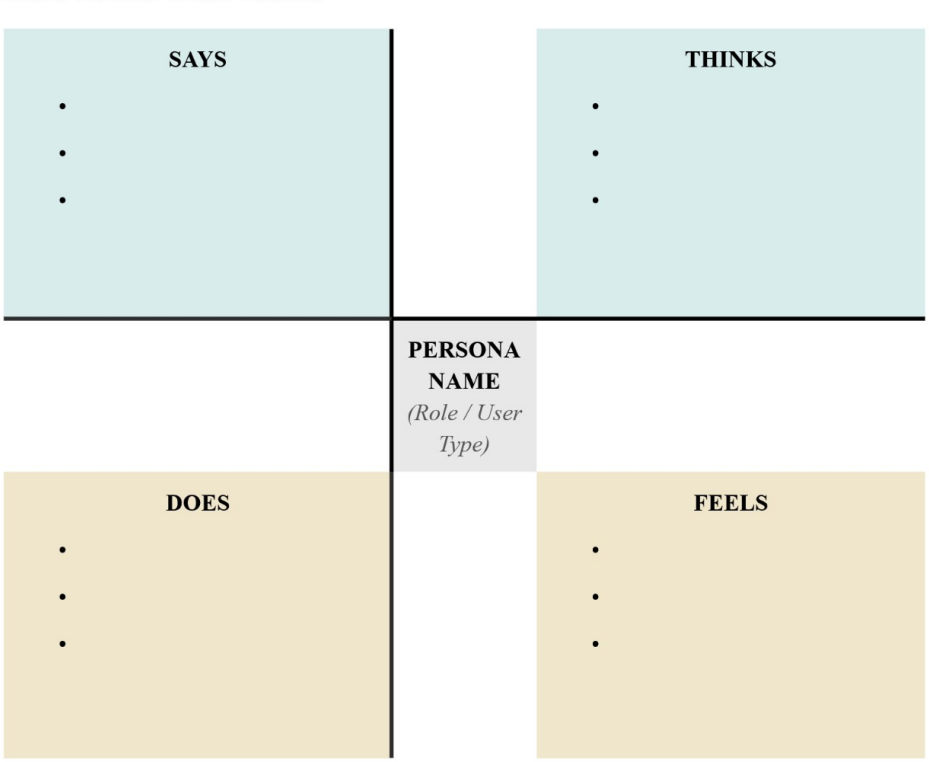
**Empathy Map Canvas:**

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes.

It is a useful tool to helps teams better understand their users.
Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise of creating the map helps participants consider things from the user's perspective along with his or her goals and challenges.

**Example:**

Reference: https://www.mural.co/templates/empathy-map-canvas




## Completed empathy canvas

Persona assumptions below are derived from the project overview, rather than customer interviews. The linked Mural template is retained as a reference attachment.

| Canvas area | Scenario-based content |
|---|---|
| Who | Emily Johnson, EV market strategy director |
| Goal and tasks | Compare sales, satisfaction, battery adoption and range to prioritize product segments. |
| Hear | Scenario assumption: leadership requests evidence for product portfolio decisions. |
| See | A large CSV with manufacturer, category, battery and country fields. |
| Say | Scenario paraphrase: I need executive-level insight into market performance. |
| Do | Compare segments, select a category and inspect the linked charts. |
| Think and feel | Needs consistent metrics and clear boundaries on synthetic-data interpretation. |

| Canvas area | Scenario-based content |
| --- | --- |
| Pains | Missing measurements, fragmented comparisons and ambiguous model-year meaning. |
| Gains | One interactive dashboard with nine views and documented calculations. |

# Completed linked Mural template

The downloaded template is filled below. The preceding editable table provides the same responses at reading size.




Scenario-derived assumptions, not interviews. Source template: Mural empathy map canvas.

# Project author

Vishwas Choyal | EN25CS3011066 | Individual project

# Customer Journey Map

| | |
|---|---|
| **Date** | 31 October 2026 |
| **Team ID** | Individual |
| **Project Name** | Electric Vehicle Market Performance and Insights |
| **Maximum Marks** | 2 Marks |

## Customer Journey Map

Map out the customer's experience stage-by-stage, capturing their actions, touchpoints, thoughts, and feelings, along with process ownership and improvement opportunities at each stage.

| Phase of journey | Discover | Compare | Decide |
|---|---|---|---|
| **Actions** *What does the customer do?* | Read project overview; open dashboard. | Select category; compare nine views. | Review evidence; note limitations. |
| **Touchpoint** *What part of the service do they interact with?* | SkillWallet overview; Tableau Public. | Category bars; chart tooltips. | Report, workbook and demo. |
| **Customer Thought** *What is the customer thinking?* | Which questions must I answer? | Are metrics comparable and complete? | What does the evidence support? |
| **Customer Feeling** *What is the customer feeling?* | Curious; uncertain about raw data. | More informed; checking assumptions. | Clearer about dataset findings. |

| | | | |
|---|---|---|---|
| **Process Ownership**<br>*Who is in the lead on this?* | Project author | Dashboard viewer | Reviewer |
| **Opportunities**<br>*How can we improve this stage?* | State synthetic scope early. | Show units and null handling. | Link evidence and reproducible files. |

# Project Design Phase-II
# Data Flow Diagram & User Stories

| | |
|---|---|
| Date | 31 October 2026 |
| Team ID | Individual |
| Project Name | Electric Vehicle Market Performance and Insights |
| Maximum Marks | 4 Marks |

**Data Flow Diagrams:**

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

**DFD Symbol Legend**

| Symbol | Name | Description |
|---|---|---|
| **Oval / Rounded shape** | **External Entity** | A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway). |
| **Rectangle with numbered header** | **Process** | An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order). |

| Symbol | Name | Description |
|---|---|---|
| **Rectangle (solid fill, no number)** | **Data Store** | A place where data is held for later use (e.g., Customer Database, Inventory). |
| **Labeled arrow** | **Data Flow** | The movement of data between entities, processes, and data stores, labeled with what data is being passed. |

### User Stories

Use the below template to list all the user stories for the product.

| User Type | Functional Requirement (Epic) | User Story Number | User Story / Task | Acceptance criteria | Priority | Release |
|---|---|---|---|---|---|---|
| Analyst | Data acquisition | US-01 | Download both Kaggle CSV splits | 300,000 source rows inventoried | High | Sprint 1 |

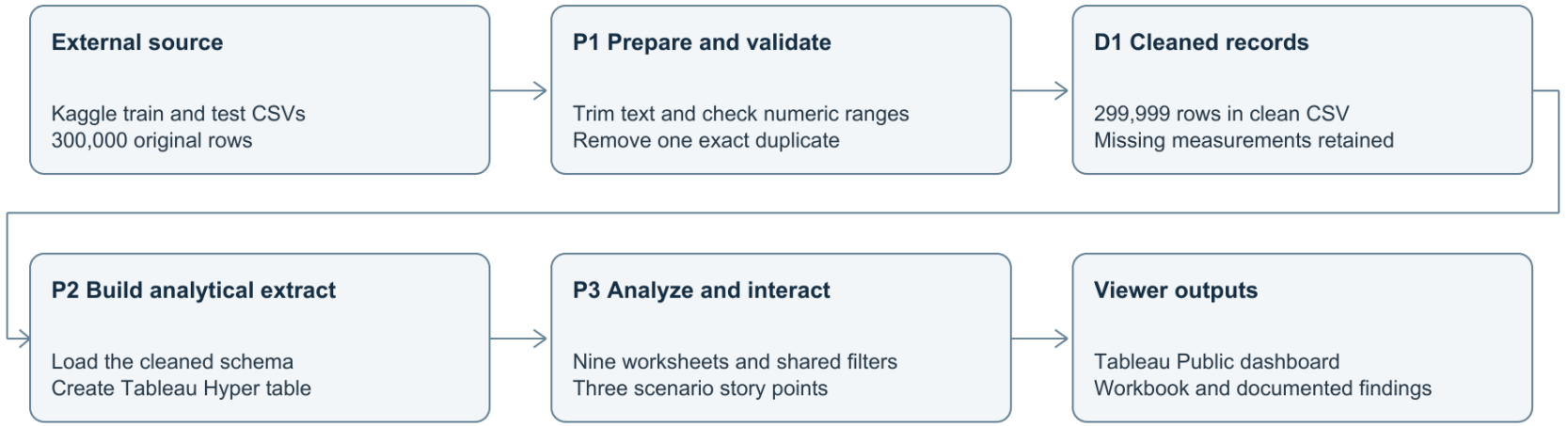
| User Type | Functional Requirement (Epic) | User Story Number | User Story / Task | Acceptance criteria | Priority | Release |
|---|---|---|---|---|---|---|
| Analyst | Data preparation | US-02 | Remove exact duplicates and preserve nulls | 299,999 rows; one duplicate removed | High | Sprint 1 |
| Emily | Market analysis | US-03 | Compare category sales and ratings | Views 1 and 3 show sums and means | High | Sprint 1 |
| Emily | Cohort analysis | US-04 | Compare model-year segment composition | Views 2 and 9 label model year correctly | High | Sprint 1 |
| David | Competition | US-05 | Compare manufacturer totals and sales shares | View 4 has ten manufacturers and share tooltips | High | Sprint 1 |
| David | Geography | US-06 | Compare manufacturing-country sales | View 6 distinguishes manufacturing from market region | High | Sprint 1 |
| Sarah | Technology | US-07 | Compare battery mix, capacity and range | Views 5, 7 and 8 cover technology metrics | High | Sprint 1 |
| Dashboard viewer | Interaction | US-08 | Select a vehicle category to filter charts | Linked charts update and clear selection restores totals | High | Sprint 1 |
| Mentor | Delivery | US-09 | Review report, workbook and narrated demo | Public links and files can be opened | High | Sprint 2 |

# Implemented data flow

Kaggle CSV files → Python validation and exact deduplication → cleaned CSV → Tableau Hyper extract → nine worksheets → interactive dashboard → Tableau Public and project documentation.

The raw files remain the source of truth. The published dashboard contains synthetic analytical data and no login credentials.

## EV project data flow




Read left to right. Upper row feeds the lower row through the rightmost data store.

# Project author

Vishwas Choyal | EN25CS3011066 | Individual project

**Project Design Phase-II**
**Solution Requirements (Functional & Non-functional)**

| | |
|---|---|
| Date | 31 October 2026 |
| Team ID | Individual |
| Project Name | Electric Vehicle Market Performance and Insights |
| Maximum Marks | 4 Marks |

**Functional Requirements:**
Following are the functional requirements of the proposed solution.

| FR No. | Functional Requirement (Epic) | Sub Requirement (Story / Sub-Task) | Priority (High/Medium/Low) |
|---|---|---|---|
| 1 | Data Access & Integration | Load both EV dataset CSV splits and retain source provenance. | High |
| 2 | Data Processing | Trim text, remove one exact duplicate and preserve missing measurements. | High |
| 3 | Business Analysis | Calculate observed sales sums, mean ratings, mean range and shares of dataset sales. | High |
| 4 | Business Insights & Analytical Outputs | Provide nine views covering all three supplied scenarios. | High |
| 5 | User Interaction & Analysis | Selecting a category filters linked views; clearing restores all categories. | High |
| 6 | Data Update & Reporting | Rebuild the clean CSV and Hyper extract when source files change; republish the workbook. | High |

**Non-functional Requirements:**
Following are the non-functional requirements of the proposed solution.

| FR No. | Non-Functional Requirement | Description |
|---|---|---|
| NFR-1 | **Usability** | Readable titles, units and consistent chart layout. |
| NFR-2 | **Security** | Publish only synthetic data and project artifacts; exclude account secrets. |
| NFR-3 | **Reliability** | Reconcile row count and total sales independently; retain null semantics. |
| NFR-4 | **Performance** | Use a Hyper extract; verify category selection updates all target charts. |
| NFR-5 | **Availability** | Portable TWBX for offline access; Tableau Public for browser access. |

| FR No. | Non-Functional Requirement | Description |
|---|---|---|
| NFR-6 | **Scalability** | Aggregate for rendering and rebuild extracts as source volume increases. |

# Project author

Vishwas Choyal | EN25CS3011066 | Individual project

# Project Design Phase-II
# Technology Stack (Architecture & Stack)

| | |
|---|---|
| Date | 31 October 2026 |
| Team ID | Individual |
| Project Name | Electric Vehicle Market Performance and Insights |
| Maximum Marks | 2 Marks |

**Define Your Technology Stack:**

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

| S.No | Component | Description | Technology | Justification / Purpose |
|---|---|---|---|---|
| 1. | Programming / Query Language | Language used to query, filter, transform, aggregate, and analyze data. | Python 3 and pandas | Repeatable validation and aggregation. |
| 2. | Database / Data Warehouse | Warehouse Used to store and manage structured business data for analysis and reporting. | CSV and Tableau Hyper | Preserve source portability and accelerate queries. |
| 3. | Data Ingestion | Used to load data from source files or external systems into the analytical environment. | Tableau text connector and Hyper API | Load the original schema and the cleaned extract. |
| 4. | Data Transformation and Business Logic | Tools/Logic used to clean, join, filter, aggregate, data for analysis and calculate KPIs, metrics, business rules, and analytical measures. | pandas and Tableau calculations | Deduplicate, preserve nulls and define analytical measures. |

| S.No | Component | Description | Technology | Justification / Purpose |
|---|---|---|---|---|
| 5. | Data Visualization / Analytics | Used to create interactive dashboards, charts, reports, KPIs, and visual analysis. | Tableau Public 2026.3 | Build nine worksheets and an interactive dashboard. |
| 6. | Front-End / User Interface | Interface through which users interact with dashboards, filters, parameters, and visual insights. | Tableau Public and GitHub | Browser access and a versioned project submission. |

## Project author

Vishwas Choyal | EN25CS3011066 | Individual project

**Project Design Phase-I**
**Problem – Solution Fit Template**

| | |
|---|---|
| Date | 31 October 2026 |
| Team ID | Individual |
| Project Name | Electric Vehicle Market Performance and Insights |
| Maximum Marks | 2 Marks |

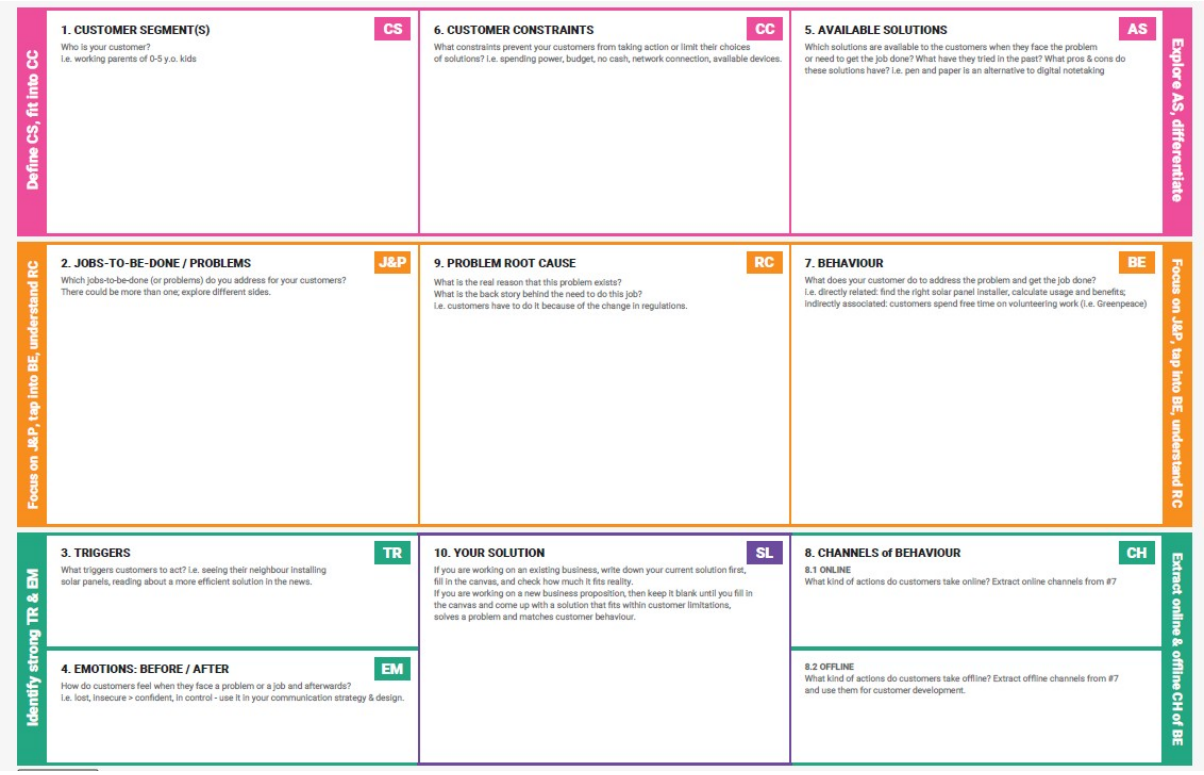
**Problem – Solution Fit Template:**

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

**Purpose:**

- ❑ Solve complex problems in a way that fits the state of your customers.
- ❑ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ❑ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ❑ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ❑ **Understand the existing situation in order to improve it for your target group.**

**Template:**




References:

1. https://www.ideahackers.network/problem-solution-fit-canvas/

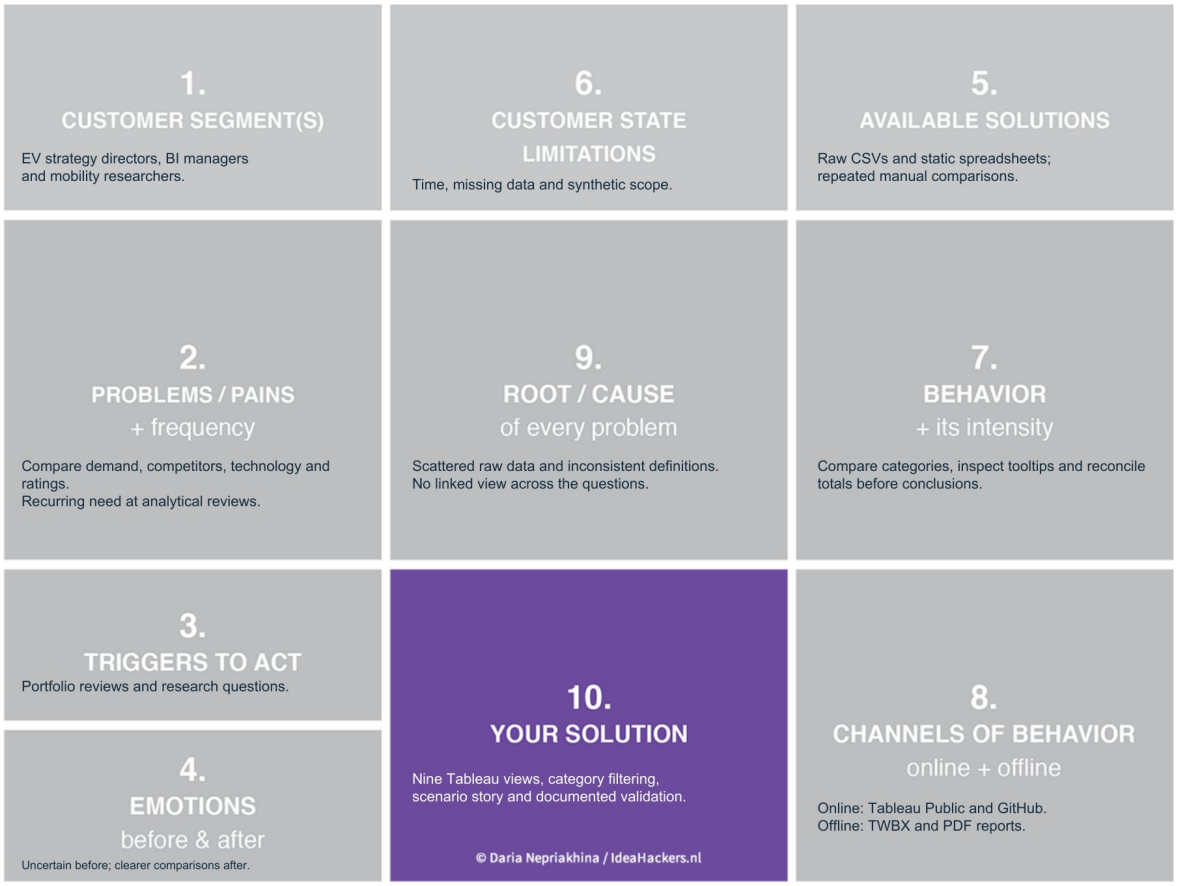
2. https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe

| Canvas area | Completed project response |
|---|---|
| Customer segments | EV strategy directors, automotive BI consultants and sustainable mobility researchers. |
| Jobs and problems | Compare segment demand, manufacturer performance, battery metrics and satisfaction. |
| Triggers | Portfolio reviews, client market-entry questions and technology research. |
| Emotions before and after | Scenario assumption: uncertainty from fragmented data; greater clarity from defined, comparable measures. |
| Available solutions | Raw CSVs and static spreadsheets; flexible but require repeated manual comparisons. |
| Constraints | Time, analytical skill, missing measurements and synthetic-data scope. |
| Behaviour | Inspect rankings, compare segments and validate totals before drawing conclusions. |
| Online channels | Tableau Public dashboard and linked project repository. |
| Offline channels | Portable TWBX and PDF documentation for review. |
| Root cause | Unstructured analytical workflow and absent shared definitions. |
| Solution | Nine-view Tableau dashboard, category filter and documented data-quality checks. |

## Completed linked IdeaHackers template

The downloaded canvas is completed below; original template attribution is retained.

**EV project problem solution fit canvas**




Scenario-derived responses. Source: IdeaHackers canvas by Daria Nepriakhina; original attribution retained.

## Project author

Vishwas Choyal | EN25CS3011066 | Individual project

SMARTBRIDGE
Let's Bridge the Gap

# Project Initialization and Planning Phase

| | |
|---|---|
| Date | 31 October 2026 |
| Team ID | Individual |
| Project Title | Electric Vehicle Market Performance and Insights |
| Maximum Marks | 3 Marks |

## Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

| | |
|---|---|
| **Project Overview** | |
| Objective | Answer the three supplied EV analysis scenarios with an interactive Tableau dashboard containing nine visualizations. |
| Scope | Descriptive analysis of 299,999 cleaned synthetic records across model years 2018–2026. No predictive model or real-market forecasts. |
| **Problem Statement** | |
| Description | Users need comparable information on segments, manufacturers, batteries, geography and customer ratings. |
| Impact | Reduce manual CSV comparisons and make assumptions and metric definitions visible. |
| **Proposed Solution** | |
| Approach | Acquire both CSVs, validate and deduplicate, build a Hyper extract, develop nine views, verify filtering and document the evidence. |
| Key Features | Category filtering, manufacturer ranking, battery comparisons, cohort heatmap and portable workbook. |

## Resource Requirements

| Resource Type | Description | Specification/Allocation |
|---|---|---|
| **Hardware** | | |
| Computing Resources | CPU/GPU specifications, number of cores | Windows PC capable of running Tableau Public; no GPU computation required. |
| Memory | RAM specifications | 8 GB minimum, 16 GB recommended. |
| Storage | Disk space for data, models, and logs | At least 2 GB available for data and deliverables. |
| **Software** | | |
| Frameworks | Python frameworks | Tableau Public desktop authoring. |
| Libraries | Additional libraries | pandas, Tableau Hyper API and document-generation libraries. |
| Development Environment | IDE, version control | Local scripts and Tableau Public; GitHub for delivery. |
| **Data** | | |
| Data | Source, size, format | Kaggle synthetic EV CSVs, 300,000 raw rows and 37 original fields. |

**Project author**
Vishwas Choyal | EN25CS3011066 | Individual project

**Project Design Phase-I**
**Solution Architecture**

| | |
|---|---|
| Date | 31 October 2026 |
| Team ID | Individual |
| Project Name | Electric Vehicle Market Performance and Insights |
| Maximum Marks | 4 Marks |

## Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Implemented component architecture

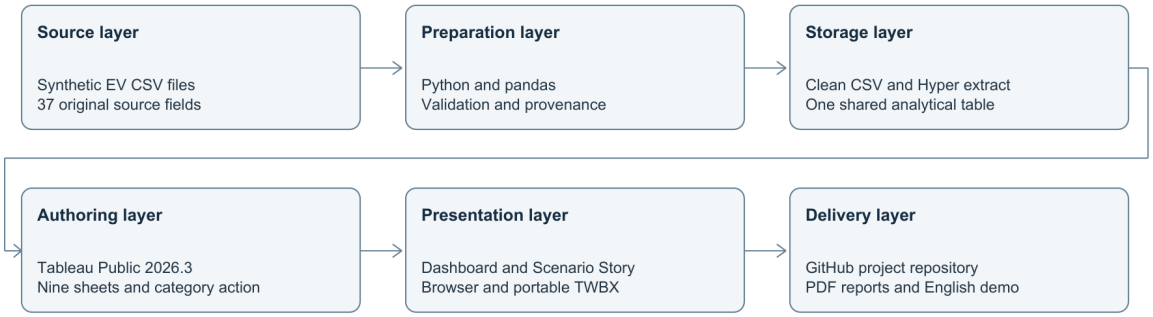
**Component Description Table**

| Component Name | Description / Role in Architecture | Technology Used |
|---|---|---|
| Source | 200,000 train and 100,000 test rows | Kaggle CSV |
| Preparation | Null audit, type checks and exact deduplication | Python and pandas |
| Analytical store | 299,999 validated records | Tableau Hyper |
| Visualization | Nine worksheets and category filter action | Tableau Public |
| Delivery | Workbook, documents and narrated demo | GitHub and Google Drive |

# Project architecture

Source files feed a repeatable preparation step. The cleaned data feeds one Hyper extract shared by all nine Tableau worksheets. The dashboard applies a category-selection filter to linked sheets and is delivered as a self-contained TWBX.

**EV dashboard solution architecture**




Read left to right. Upper row feeds the lower row through the rightmost data store.

# Project author

Vishwas Choyal | EN25CS3011066 | Individual project

# Project Planning Phase

## Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

| | |
|---|---|
| Date | 31 October 2026 |
| Team ID | Individual |
| Project Name | Electric Vehicle Market Performance and Insights |
| Maximum Marks | 8 Marks |

### Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

| Sprint | Functional Requirement (Epic) | User Story Number | User Story / Task | Story Points | Priority | Team Members |
|---|---|---|---|---|---|---|
| Sprint 1 | Acquisition | US-01 | Download source and inspect schema | 2 | High | Vishwas Choyal |
| Sprint 1 | Preparation | US-02 | Validate and deduplicate records | 3 | High | Vishwas Choyal |
| Sprint 1 | Visualization | US-03 | Build nine charts | 5 | High | Vishwas Choyal |
| Sprint 1 | Dashboard | US-04 | Verify category filtering | 3 | High | Vishwas Choyal |
| Sprint 1 | Testing | US-05 | Reconcile totals and verify portable file | 2 | High | Vishwas Choyal |

| Sprint | Functional Requirement (Epic) | User Story Number | User Story / Task | Story Points | Priority | Team Members |
|---|---|---|---|---|---|---|
| Sprint 2 | Documentation | US-06 | Complete phase reports | 3 | High | Vishwas Choyal |
| Sprint 2 | Demonstration | US-07 | Record English narrated walkthrough | 3 | High | Vishwas Choyal |
| Sprint 2 | Submission | US-08 | Publish links and submit for mentor review | 2 | High | Vishwas Choyal |

**Project Tracker, Velocity & Burndown Chart: (4 Marks)**

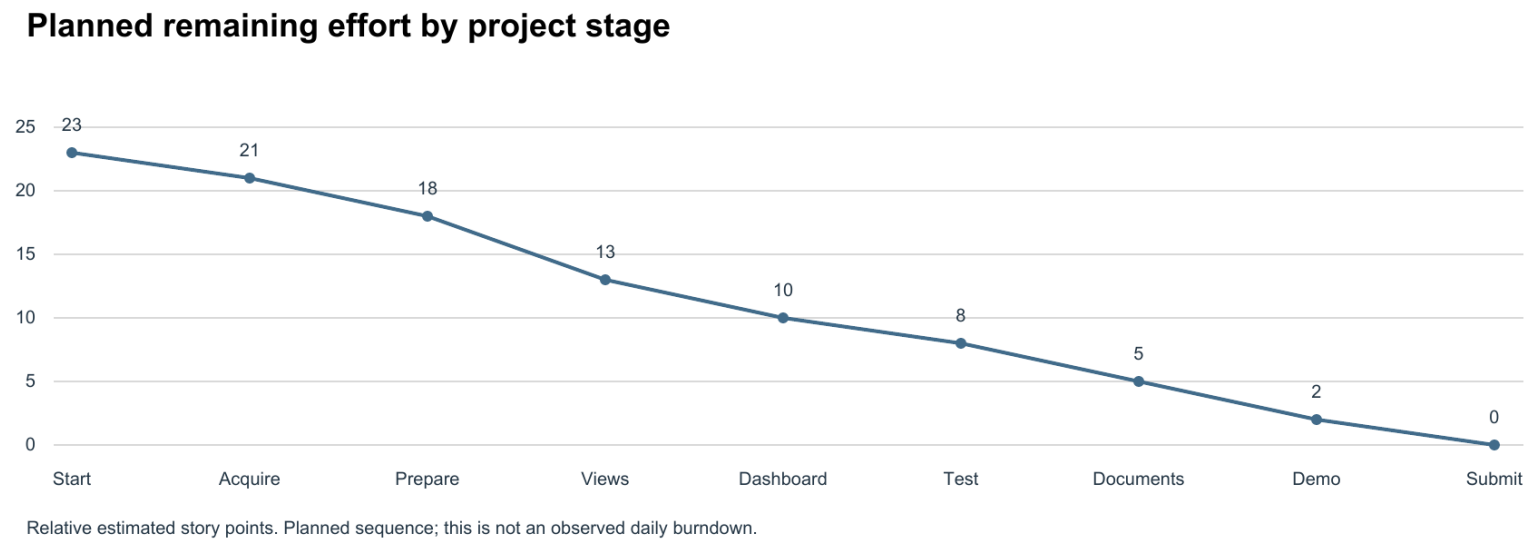
| Sprint | Total Story Points | Duration | Sprint Start Date | Sprint End Date (Planned) | Story Points Completed (as on Planned End Date) | Sprint Release Date (Actual) |
|---|---|---|---|---|---|---|
| Sprint 1 | 15 | 1 day | 31 Oct 2026 | 31 Oct 2026 | 15 | 31 Oct 2026 |
| Sprint 2 | 8 | 1 day | 31 Oct 2026 | 01 Nov 2026 | 3 | In progress |

# Planning assumptions

Story points are relative estimates for this individual capstone. Historical team velocity is not available. The tracker is a snapshot after completion of the narrated demo on 1 November: 18 of 23 estimated points completed, with documentation checks and submission in progress.

# Planned burndown

The chart shows estimated remaining points by planned stage, rather than measured daily productivity. Final review is outside the development effort estimate.




Relative estimated story points. Planned sequence; this is not an observed daily burndown.

# Project author

Vishwas Choyal | EN25CS3011066 | Individual project

| | |
|---|---|
| Date | 31 October 2026 |
| Team ID | Individual |
| Project Name | Electric Vehicle Market Performance and Insights |
| Maximum Marks | 5 Marks |

Track applicability: This project uses Tableau Public and a local Hyper extract. Snowflake SQL, ingestion and query-history evidence are not applicable. See the corresponding Tableau reports for implemented work.

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting

**Business Questions and Visualisation**

The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations using data stored in Snowflake. Well-framed business questions help identify the required metrics, determine the appropriate Snowflake tables or analytical views, select relevant data, and create visualizations that provide useful business insights.

## Project author

Vishwas Choyal | EN25CS3011066 | Individual project

# Snowflake Project – Data Engineering & SQL Solution

| | |
|---|---|
| Date | 31 October 2026 |
| Team ID | Individual |
| Project Name | Electric Vehicle Market Performance and Insights |
| Maximum Marks | 5 Marks |

Track applicability: This project uses Tableau Public and a local Hyper extract. Snowflake SQL, ingestion and query-history evidence are not applicable. See the corresponding Tableau reports for implemented work.

This section evaluates the quality and completeness of the implemented Snowflake solution. The submission should demonstrate that the data has been prepared, cleaned, transformed, validated, and made ready for downstream analysis..

**Instructions**

1. Ensure SQL scripts are well-structured, clearly organized, and follow appropriate SQL practices.
2. The solution should address the project requirements through the required data preparation and transformation steps.
3. Include all SQL scripts, table/view definitions, and setup information needed to reproduce the solution.
4. Provide evidence that the final Snowflake data is accurate, consistent, and ready for analysis.

**Solution Summary**

| Field | Details |
|---|---|
| Project / Repository Link | https://github.com/VishwasChoyal0/ev-market-insights-tableau |
| Data Source(s) Link | Kaggle synthetic EV dataset; source link is provided in the final report. |
| Database | Not applicable to Tableau track |
| Warehouse | Not applicable to Tableau track |

| Field | Details |
|---|---|
| Tables / Views Created | Not applicable to Tableau track |
| Key Transformations / Calculations | Not applicable to Tableau track |
| KPIs created | Not applicable to Tableau track |
| Key Insights | Not applicable to Tableau track |

**Snowflake Quality Checklist**

| S.No | Criteria | Status (Yes / No) |
|---|---|---|
| 1 | Required data is successfully loaded and accessible in Snowflake | Not applicable to Tableau track |
| 2 | Data is cleaned, transformed, and prepared appropriately | Not applicable to Tableau track |
| 3 | Required tables, views, and data relationships are correctly implemented | Not applicable to Tableau track |
| 4 | Required SQL logic, calculations, and KPIs are correctly implemented | Not applicable to Tableau track |
| 5 | Data and solution outputs are validated against the project requirements | Not applicable to Tableau track |
| 6 | Solution is properly organized, documented, and functional | Not applicable to Tableau track |

Additional Notes / Comments:

## Project author

Vishwas Choyal | EN25CS3011066 | Individual project

| | |
|---|---|
| Date | 31 October 2026 |
| Team ID | Individual |
| Project Name | Electric Vehicle Market Performance and Insights |
| Maximum Marks | 5 Marks |

Track applicability: This project uses Tableau Public and a local Hyper extract. Snowflake SQL, ingestion and query-history evidence are not applicable. See the corresponding Tableau reports for implemented work.

### Activity 1: Interactive and visually appealing dashboards

Creating an effective Streamlit dashboard involves presenting business data stored in Snowflake through clear, interactive, and easy-to-understand visualizations. Students should design the dashboard so that users can quickly understand important KPIs, trends, comparisons, and business insights.

- Clear and Intuitive Layout
- Use Appropriate Visualizations
- Colour and Theming
- Interactive Filters and Slicers
- Drill-Down Capabilities
- Responsive Design
- Custom Visuals and Icons
- Use of Infographics

## Project author

Vishwas Choyal | EN25CS3011066 | Individual project

| | |
|---|---|
| Date | 31 October 2026 |
| Team ID | Individual |
| Project Name | Electric Vehicle Market Performance and Insights |
| Maximum Marks | 5 Marks |

Track applicability: This project uses Tableau Public and a local Hyper extract. Snowflake SQL, ingestion and query-history evidence are not applicable. See the corresponding Tableau reports for implemented work.

Data preparation refers to the process of cleaning, standardizing, transforming, and organizing data so that it is suitable for analysis and reporting. In a Snowflake project, data preparation involves working with the ingested RAW data, identifying data quality issues, creating STAGING tables, applying required transformations, and preparing reliable data for the CURATED and MART layers.

**Activity 1: Identifying and Handling Null Values**

Identify the NULL or missing values present in the Snowflake dataset, analyze the affected columns, determine the appropriate approach for handling them based on the data and business requirements, apply the required query, and validate the cleaned data to ensure it is ready for further analysis and visualization.

Not applicable: Snowflake was not used. See the Tableau preparation and testing reports.

**Activity 2: Duplicate Record Handling**

Identify and analyze duplicate records in the Snowflake dataset, determine whether the duplicates are genuine or repeated entries, define the appropriate criteria for identifying duplicates, remove or retain records based on the project requirements, and validate the dataset after treatment to ensure that duplicate records do not affect the accuracy of further analysis and visualizations.

Not applicable: Snowflake was not used. See the Tableau preparation and testing reports.

**Activity 3: Data Cleaning and Standardization**

Identify and correct inconsistencies, invalid entries, formatting issues, and data quality problems in the Snowflake dataset, standardize values and formats across relevant columns, apply the required SQL-based cleaning and transformation steps, and validate the cleaned and standardized dataset to ensure consistency, accuracy, and readiness for further analysis and visualization.

Not applicable: Snowflake was not used. See the Tableau preparation and testing reports.

**Activity 4: Data Transformation and Validation**

Apply the required data transformations to prepare the cleaned data for analysis and modelling. Create necessary derived fields, apply business rules, filter or transform records where required, and validate the prepared data and other relevant quality checks before using it for the CURATED and MART layers.

Not applicable: Snowflake was not used. See the Tableau preparation and testing reports.

## Project author

Vishwas Choyal | EN25CS3011066 | Individual project

# Business Question and Visualization Report

| | |
|---|---|
| Date | 31 October 2026 |
| Team ID | Individual |
| Project Name | Electric Vehicle Market Performance and Insights |
| Maximum Marks | 5 Marks |

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting

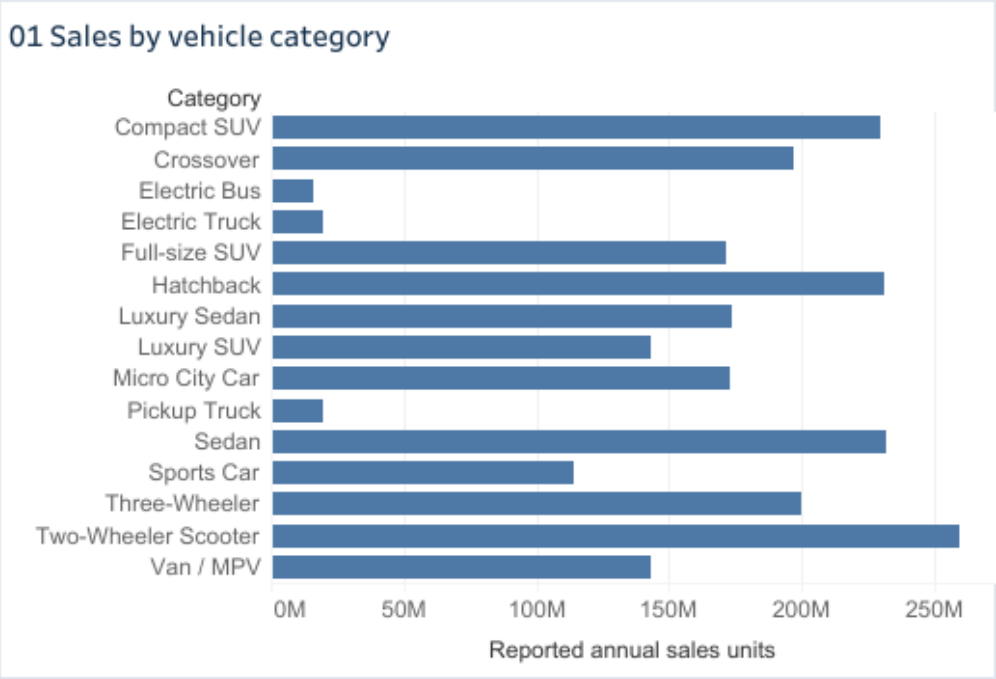
## Business Questions and Visualisation

The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations in Tableau. Well-framed questions help in identifying key metrics, selecting relevant data, and building visualisation that provide insights.

# Visualization 1

Which vehicle categories contribute most reported sales?

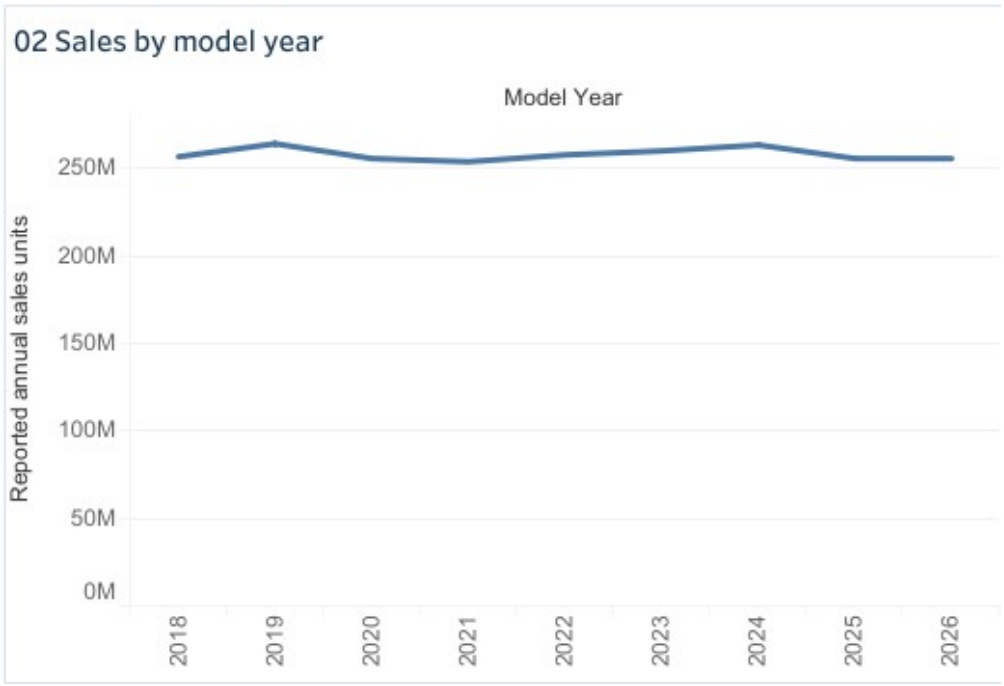
Visual: Bar chart. Two-Wheeler Scooter contributes the largest reported category sales, 259,647,705 units (11.2% of dataset sales).



# Visualization 2

How do reported sales differ across model-year cohorts?

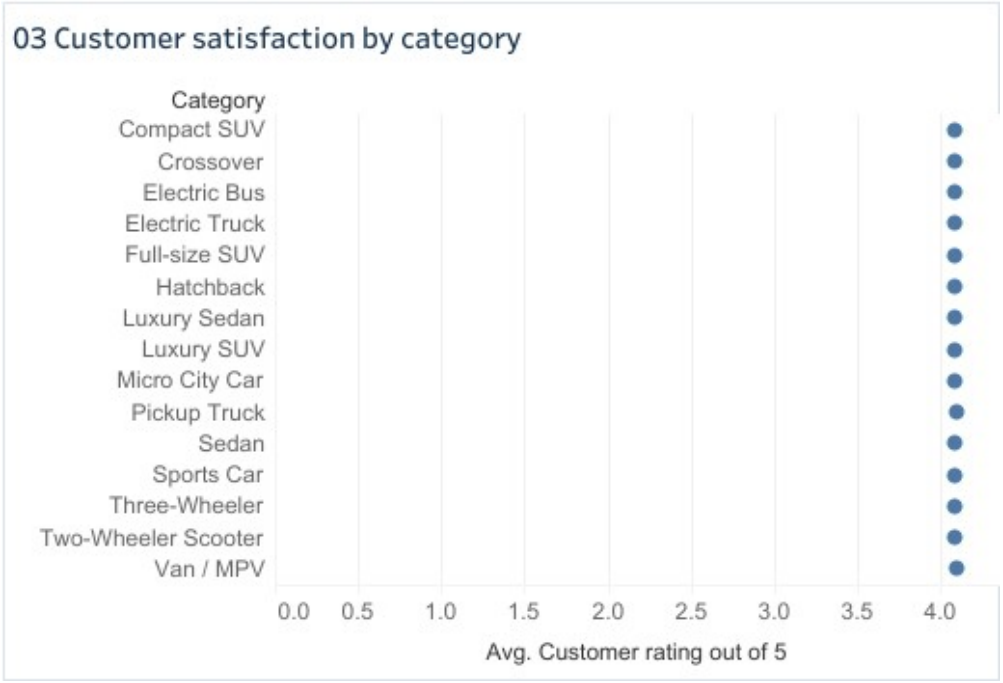
Visual: Line chart. Model-year cohorts span 2018 to 2026. Differences between cohorts are not evidence of transaction-year market growth.



# Visualization 3

How do average customer ratings differ across vehicle categories?

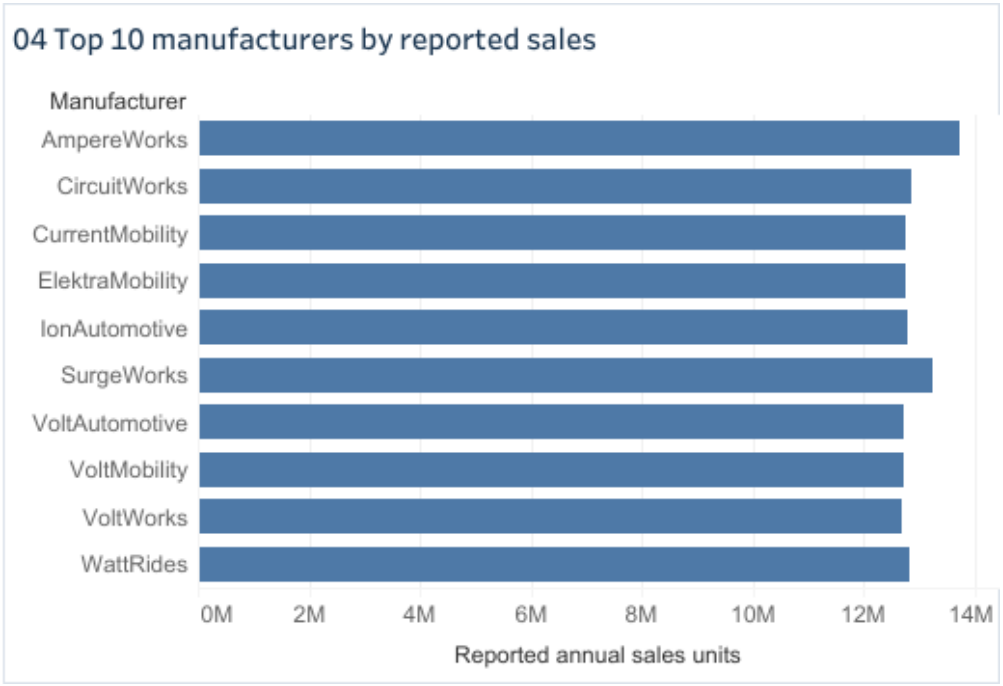
Visual: Circle chart. Pickup Truck has the highest mean category rating, 4.09 out of five, among records with observed ratings.



# Visualization 4

Which ten manufacturers lead reported sales and what shares do they contribute?

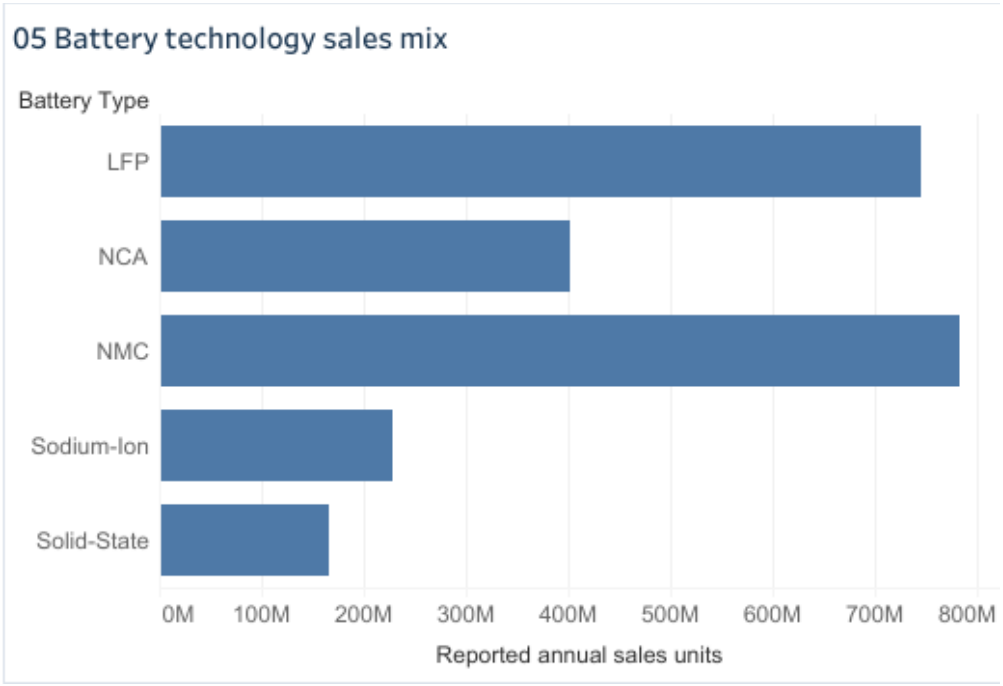
Visual: Bar chart. AmpereWorks leads reported manufacturer sales with 13,736,241 units (0.59% of dataset sales). Names are fictional.




# Visualization 5

Which battery technologies account for the most reported sales?

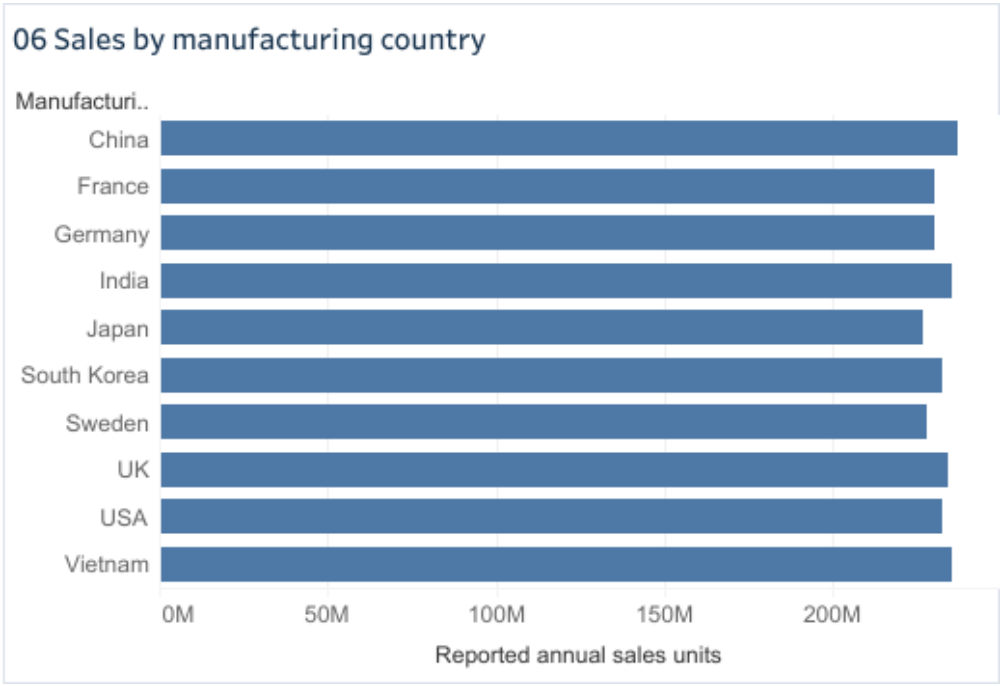
Visual: Bar chart. NMC leads battery-technology sales with 33.7% of reported dataset sales.



# Visualization 6

Which manufacturing countries are associated with the most reported sales?

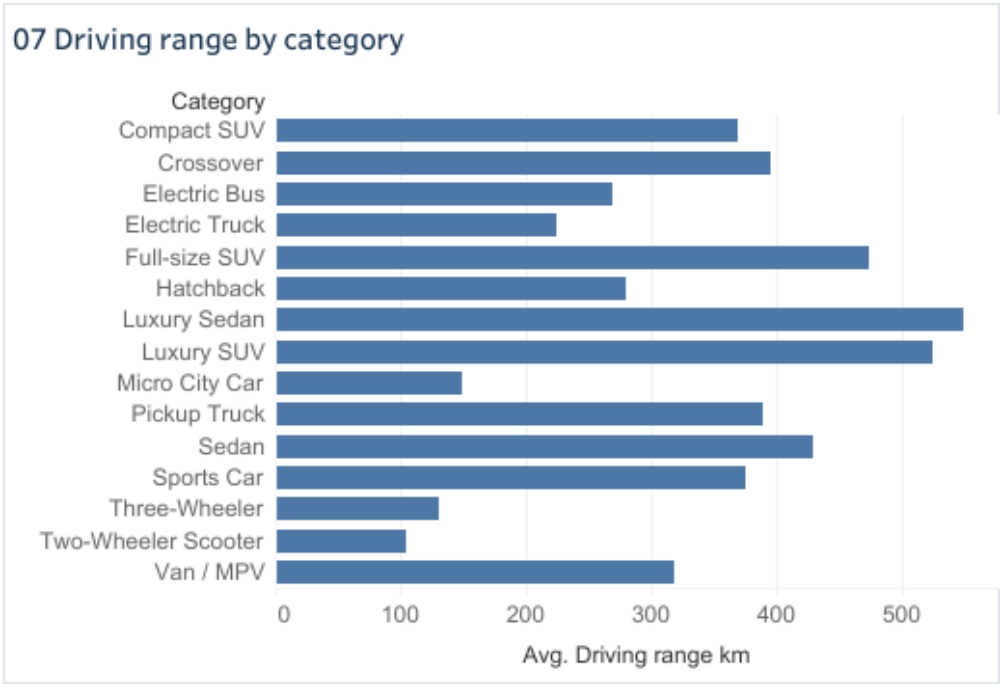
Visual: Bar chart. China leads sales associated with manufacturing country at 237,384,272 units. This measures neither factory output nor destination sales.



# Visualization 7

How does average driving range differ across vehicle categories?

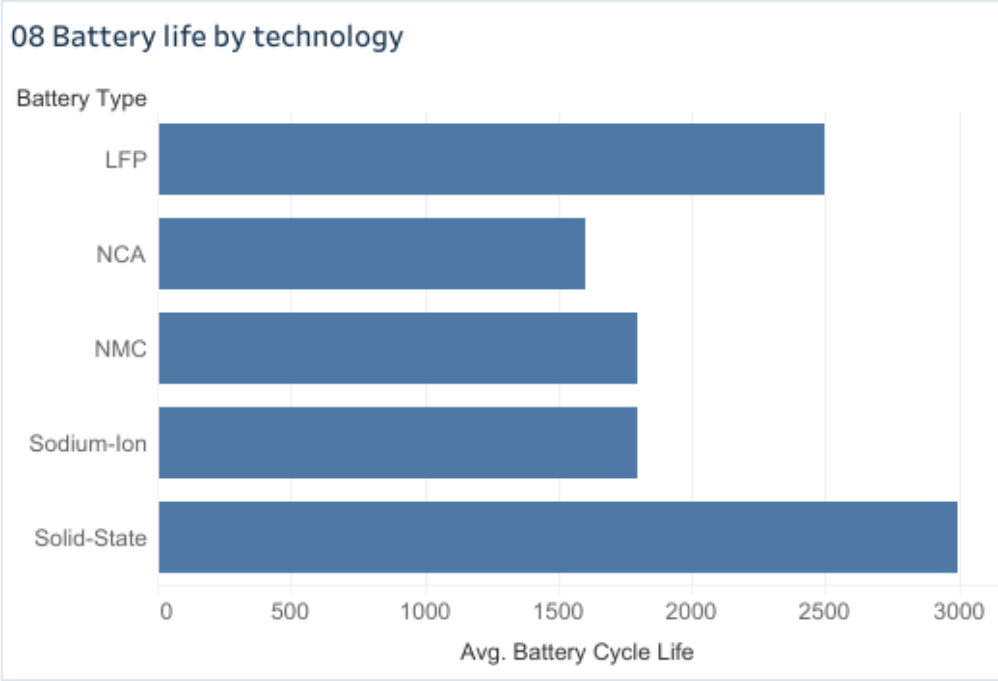
Visual: Bar chart. Luxury Sedan has the highest mean category range, 550.2 km. Different vehicle use cases limit direct comparisons.



# Visualization 8

How do battery technologies compare in average cycle life and driving range?

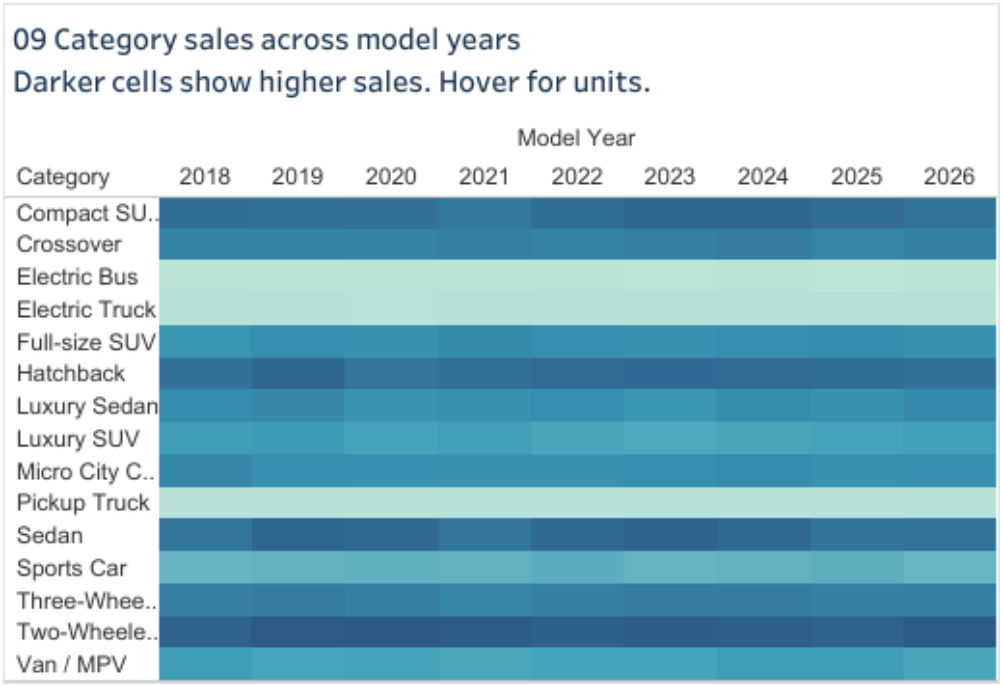
Visual: Bar chart. Average battery cycle life varies by chemistry. Compare cycle life in the bars and inspect average range and capacity in tooltips. These are synthetic technology characteristics.



# Visualization 9

How does the category sales mix differ across model-year cohorts?

Visual: Square chart. The category-by-model-year heatmap highlights large sales cells and supports cohort composition comparisons. Missing sales remain unknown.




# Interpretation limits

The publisher identifies this as synthetic data with fictional manufacturers. Sales by model year describe cohorts, and average ratings are unweighted source-record means.

# Project author

Vishwas Choyal | EN25CS3011066 | Individual project

| | |
|---|---|
| Date | 31 October 2026 |
| Team ID | Individual |
| Project Name | Electric Vehicle Market Performance and Insights |
| Maximum Marks | 5 Marks |

Creating an effective dashboard involves thoughtful design to ensure that the presented information is clear, relevant, and easily understandable for the intended audience. Here are some key principles and best practices for dashboard design

**Activity 1: Interactive and visually appealing dashboards**

Creating interactive and visually appealing dashboards involves a combination of thoughtful design, effective use of visual elements, and the incorporation of interactive features. Here are some tips to help you design dashboards that are both visually appealing and engaging for users so take care of below points

- Clear and Intuitive Layout
- Use Appropriate Visualizations
- Colour and Theming
- Interactive Filters and Slicers
- Drill-Down Capabilities
- Responsive Design
- Custom Visuals and Icons
- Use of Infographics

# Implemented dashboard

A 1600 by 1200 canvas arranges nine views in three rows. The first chart is a category-selection filter. Tooltips expose ratings, sales and record counts. All chart panels use Entire View fitting.

The title identifies the synthetic data and record count. Model-year labels avoid implying transaction dates.

Two-Wheeler Scooter contributes the largest reported category sales, 259,647,705 units (11.2% of dataset sales).

Model-year cohorts span 2018 to 2026. Differences between cohorts are not evidence of transaction-year market growth.

Pickup Truck has the highest mean category rating, 4.09 out of five, among records with observed ratings.

AmpereWorks leads reported manufacturer sales with 13,736,241 units (0.59% of dataset sales). Names are fictional.

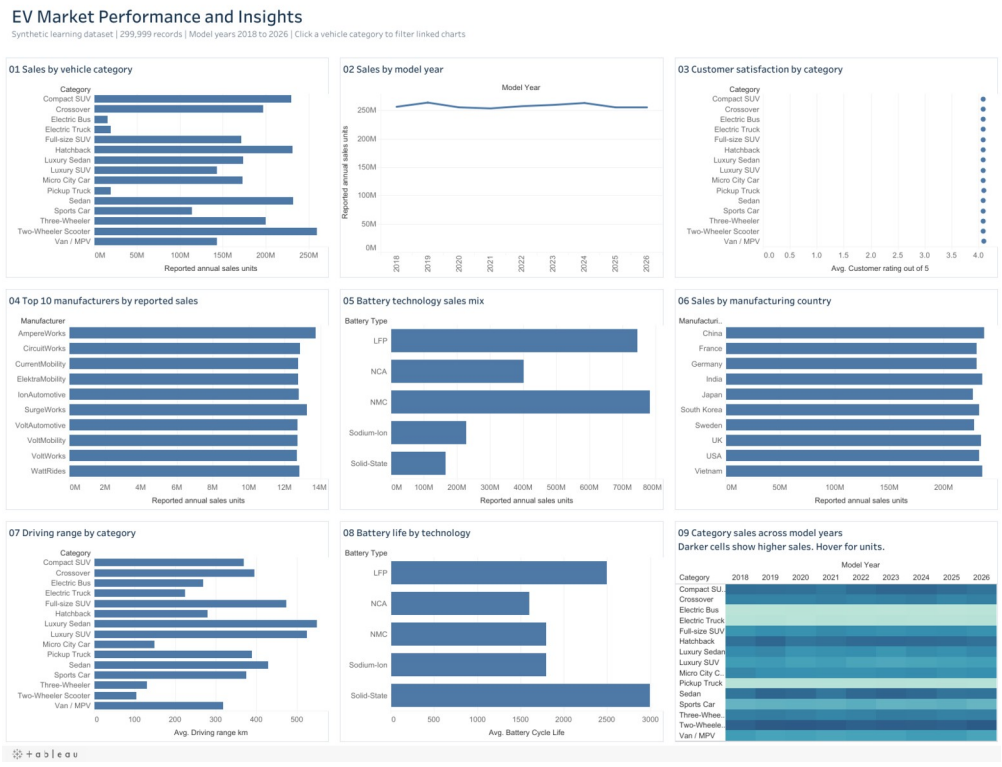
NMC leads battery-technology sales with 33.7% of reported dataset sales.

China leads sales associated with manufacturing country at 237,384,272 units. This measures neither factory output nor destination sales.

Luxury Sedan has the highest mean category range, 550.2 km. Different vehicle use cases limit direct comparisons.

Average battery cycle life varies by chemistry. Compare cycle life in the bars and inspect average range and capacity in tooltips. These are synthetic technology characteristics.

The category-by-model-year heatmap highlights large sales cells and supports cohort composition comparisons. Missing sales remain unknown.




# Public dashboard

https://public.tableau.com/views/EVMarketInsights_17934697727980/EVMarketInsights

# Project author

Vishwas Choyal | EN25CS3011066 | Individual project

| | |
|---|---|
| Date | 31 October 2026 |
| Team ID | Individual |
| Project Name | Electric Vehicle Market Performance and Insights |
| Maximum Marks | 5 Marks |

By using stories in Tableau, you can effectively communicate complex data in a way that is both interactive and engaging, making it easier for the audience to follow along and understand the insights. It's a tool for **data storytelling**, allowing you to present insights in a cohesive, engaging way that takes viewers through a logical progression of findings or analyses.

In Tableau, **Story** is a feature that allows you to create a sequence of dashboards, visualizations, and text to present data insights in a cohesive and narrative-driven way. It's like a slideshow within Tableau that guides the audience through a series of data points, helping them understand key insights, trends, or outcomes of your analysis.

# Scenario narrative

Story point 1 introduces Emily through category sales. The separate dashboard adds satisfaction, driving range and model-year cohort comparisons.

Story point 2 introduces David through the top manufacturers. Dashboard tooltips and linked views add sales contributions, battery mix and manufacturing countries.

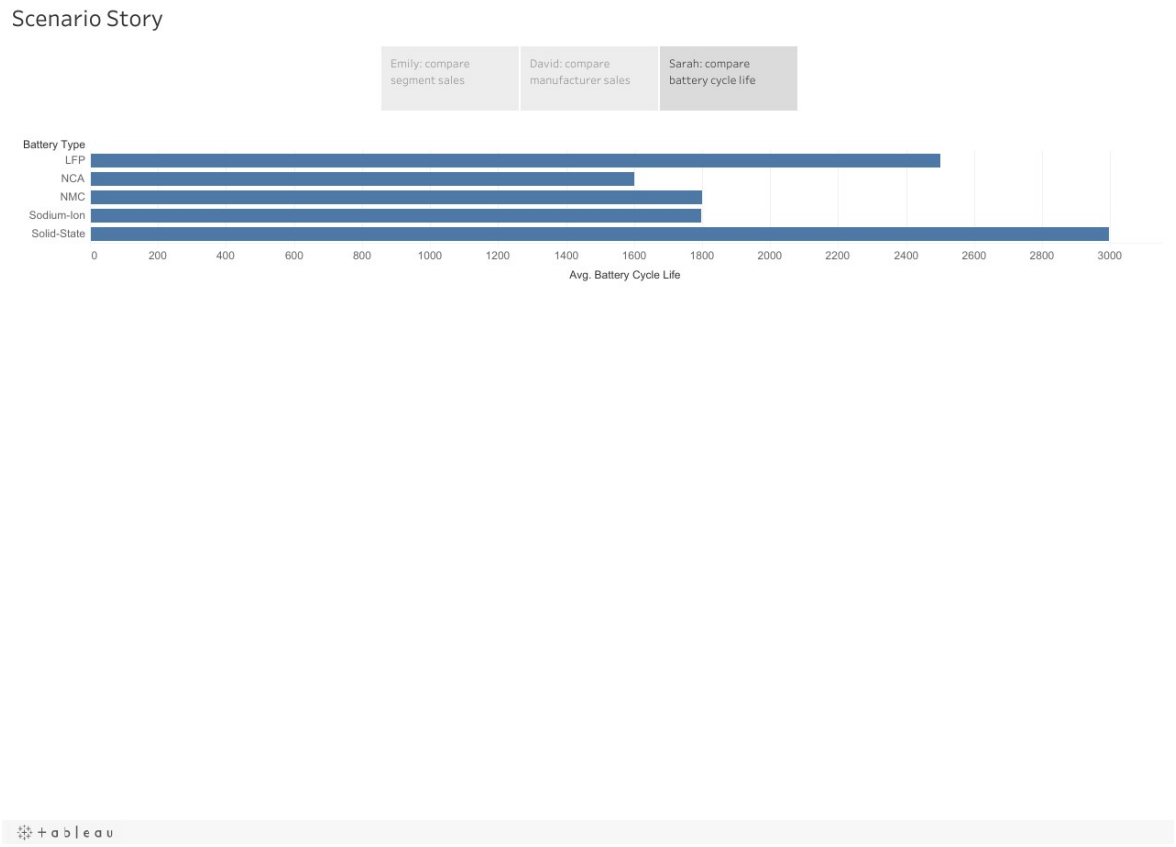
Story point 3 introduces Sarah through battery cycle life. Tooltips add average range and capacity; the separate dashboard adds category and geography comparisons.

# Narrative conclusion

The dashboard connects commercial, technical and customer measures. The synthetic data supports an analytical demonstration, while real decisions would require measured market data and transaction dates.

# Story publication

https://public.tableau.com/views/EVMarketInsights_17934697727980/ScenarioStory




# Project author

Vishwas Choyal | EN25CS3011066 | Individual project

## Performance Test

| | |
|---|---|
| Date | 31 October 2026 |
| Team ID | Individual |
| Project Name | Electric Vehicle Market Performance and Insights |
| Maximum Marks | |

Track applicability: This project uses Tableau Public and a local Hyper extract. Snowflake SQL, ingestion and query-history evidence are not applicable. See the corresponding Tableau reports for implemented work.

**Project Performance Testing:**

The project team shall fill the following information in the Project performance testing template.

| S.No. | Parameter | Screenshot / Values |
|---|---|---|
| 1 | Data Rendered | Not applicable to Tableau track |
| 2 | Data Preprocessing & Quality | Not applicable to Tableau track |
| 3. | New Fields Created | Not applicable to Tableau track |
| 4. | Filter Efficiency Test | Not applicable to Tableau track |
| 5. | Dashboard & Visualization Testing | Not applicable to Tableau track |
| 6 | Query History | Not applicable to Tableau track |

# Project author

Vishwas Choyal | EN25CS3011066 | Individual project

**Project Development Phase**
**Performance Test**

| | |
|---|---|
| Date | 31 October 2026 |
| Team ID | Individual |
| Project Name | Electric Vehicle Market Performance and Insights |
| Maximum Marks | |

**Project Performance Testing:**

Project team shall fill the following information in model performance testing template.

| S.No. | Parameter | Screenshot / Values |
|---|---|---|
| 1 | Data Rendered | 299,999 cleaned rows; 37 source fields plus provenance and record key. Native Tableau load verified. |
| 2 | Data Preprocessing | One exact duplicate removed. Missing sales and ratings remain null; no invalid values in the selected range checks. |
| 3 | Utilization of Filters | Hatchback selection changed trend, battery and manufacturer charts; clearing restored full data. |
| 4 | Calculation fields Used | Record Count = 1. Sales Share uses SUM(sales) divided by the fixed overall dataset sum. |
| 5 | Dashboard design | 9 visualizations. 1600 by 1200 dashboard with Entire View panel fitting. |
| 6 | Story Design | Three scenario-based narrative scenes; native story status is recorded in final documentation. |

# Reconciliation

Independent Python total: 2,323,303,050 reported sales units. Hatchback tooltip: 231,033,059 units and 25,713 records.
Filter behavior was observed in Tableau. No dashboard latency measurement is claimed; preparation runtime alone is not an interaction-performance test.

# Project author

Vishwas Choyal | EN25CS3011066 | Individual project

# Communication

| | |
|---|---|
| **Date** | 31 October 2026 |
| **Team ID** | Individual |
| **Project Name** | Electric Vehicle Market Performance and Insights |
| **Maximum Marks** | 1 Mark |

### Communication Plan:

Effective communication is essential for a successful project demonstration. This document outlines the communication strategy used within the team and with stakeholders throughout the project lifecycle, including how updates, issues, and feedback were managed.

| S.No | Communication Type | Frequency | Channel / Tool | Participants | Purpose |
|---|---|---|---|---|---|
| 1 | Solo check-in | Daily plan | Local checklist | Vishwas Choyal | Track tasks |
| 2 | Progress review | Per milestone | Project notes | Vishwas Choyal | Check evidence |
| 3 | Issue review | As needed | Validation log | Vishwas Choyal | Resolve defects |
| 4 | Mentor review | At submission | SkillWallet | Author; mentor | Request review |
| 5 | Demo rehearsal | Before recording | Tableau; script | Vishwas Choyal | Check clarity |
| 6 | Final handoff | Once | GitHub; report | Author; mentor | Share outputs |

### Communication Challenges & Resolutions:

| S.No | Challenge Faced | Resolution / Action Taken |
|---|---|---|
| 1 | Missing source values | Keep nulls; document coverage. |
| 2 | Model year ambiguity | Describe cohorts, not transaction years. |
| 3 | Synthetic data scope | Label fictional data throughout. |

# Demonstration of Proposed Features

| | |
|---|---|
| **Date** | 31 October 2026 |
| **Team ID** | Individual |
| **Project Name** | Electric Vehicle Market Performance and Insights |
| **Maximum Marks** | 1 Mark |

### Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

| S.No | Feature Name | Description | Status (Implemented / Partial / Pending) | Demonstrated (Yes / No) | Remarks |
|---|---|---|---|---|---|
| 1 | Source preparation | Deduplicate CSVs | Implemented | Yes | 299,999 rows |
| 2 | Nine charts | Scenario coverage | Implemented | Yes | Views 1 to 9 |
| 3 | Category filter | Linked exploration | Implemented | Yes | Reset verified |
| 4 | Portable workbook | Embedded extract | Implemented | Yes | TWBX package |
| 5 | Public dashboard | Browser access | Implemented | Yes | Tableau Public |
| 6 | Narrated demo | English walkthrough | Implemented | Yes | See report link |

### Feature Implementation Summary:

| | |
|---|---|
| **Total Features Proposed** | 6 |
| Total Features Implemented | 6 |
| Total Features Demonstrated | 6 |
| Overall Implementation Rate (%) | 100% |

# Project Demo Planning

| | |
|---|---|
| **Date** | 31 October 2026 |
| **Team ID** | Individual |
| **Project Name** | Electric Vehicle Market Performance and Insights |
| **Maximum Marks** | 1 Mark |

### Project Demo Planning:

A well-structured demo plan ensures that the team presents the project effectively, covering all key aspects in a clear and organized manner. This document outlines the plan for demonstrating the project, including the flow of the demo, key features to highlight, and responsibilities of each team member.

| S.No | Demo Section | Description | Duration (mins) | Responsible Member |
|---|---|---|---|---|
| 1 | Problem and scope | Three scenarios; synthetic data | 0.6 | Vishwas Choyal |
| 2 | Data preparation | Rows, duplicate and missing values | 0.6 | Vishwas Choyal |
| 3 | Market strategy | Views 1, 2, 3 and 9 | 2.1 | Vishwas Choyal |
| 4 | Competitive analysis | Views 4, 5 and 6 | 1.5 | Vishwas Choyal |
| 5 | Technology research | Views 7 and 8; tooltips | 1.1 | Vishwas Choyal |
| 6 | Interaction and conclusion | Category filter; limits; links | 0.7 | Vishwas Choyal |

### Demo Flow Summary:

| Step | Activity | Notes |
|---|---|---|
| 1 | Introduction & Problem Statement | Introduce the project and personas. |
| 2 | Solution Overview | Explain data preparation and nine views. |
| 3 | Live Feature Demonstration | Walk through charts and filter action. |
| 4 | Q&A Session | Discuss limits and potential improvements. |

# Scalability & Future Plan

| | |
|---|---|
| **Date** | 31 October 2026 |
| **Team ID** | Individual |
| **Project Name** | Electric Vehicle Market Performance and Insights |
| **Maximum Marks** | 1 Mark |

### Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

### Current System Limitations:

| S.No | Limitation | Impact | Priority to Address (High / Medium / Low) |
|---|---|---|---|
| 1 | Synthetic observations | No real-market inference | High |
| 2 | No transaction dates | No verified sales growth | High |
| 3 | Static extract | Manual refresh required | Medium |

### Scalability Plan:

| S.No | Scalability Aspect | Current State | Proposed Upgrade / Solution |
|---|---|---|---|
| 1 | User load | Public browser view | Monitor load; tune views |
| 2 | Data storage | Local Hyper extract | Incremental refresh pipeline |
| 3 | Performance | Aggregated chart marks | Benchmark larger extracts |
| 4 | Security | Synthetic public dataset | Private deployment for real data |

### Future Roadmap:

| Phase | Planned Feature / Enhancement | Target Timeline | Expected Impact |
|---|---|---|---|

| | | | |
|---|---|---|---|
| Phase 2 | Validate with measured market data | After source approval | Real-market evidence |
| Phase 3 | Add dated sales and refresh automation | After date fields exist | Verified temporal analysis |
| Phase 4 | Compare battery metrics within category | After validation | Reduce mix-related bias |

# Team Involvement in Demonstration

| | |
|---|---|
| **Date** | 31 October 2026 |
| **Team ID** | Individual |
| **Project Name** | Electric Vehicle Market Performance and Insights |
| **Maximum Marks** | 1 Mark |

### Team Involvement in Demonstration:

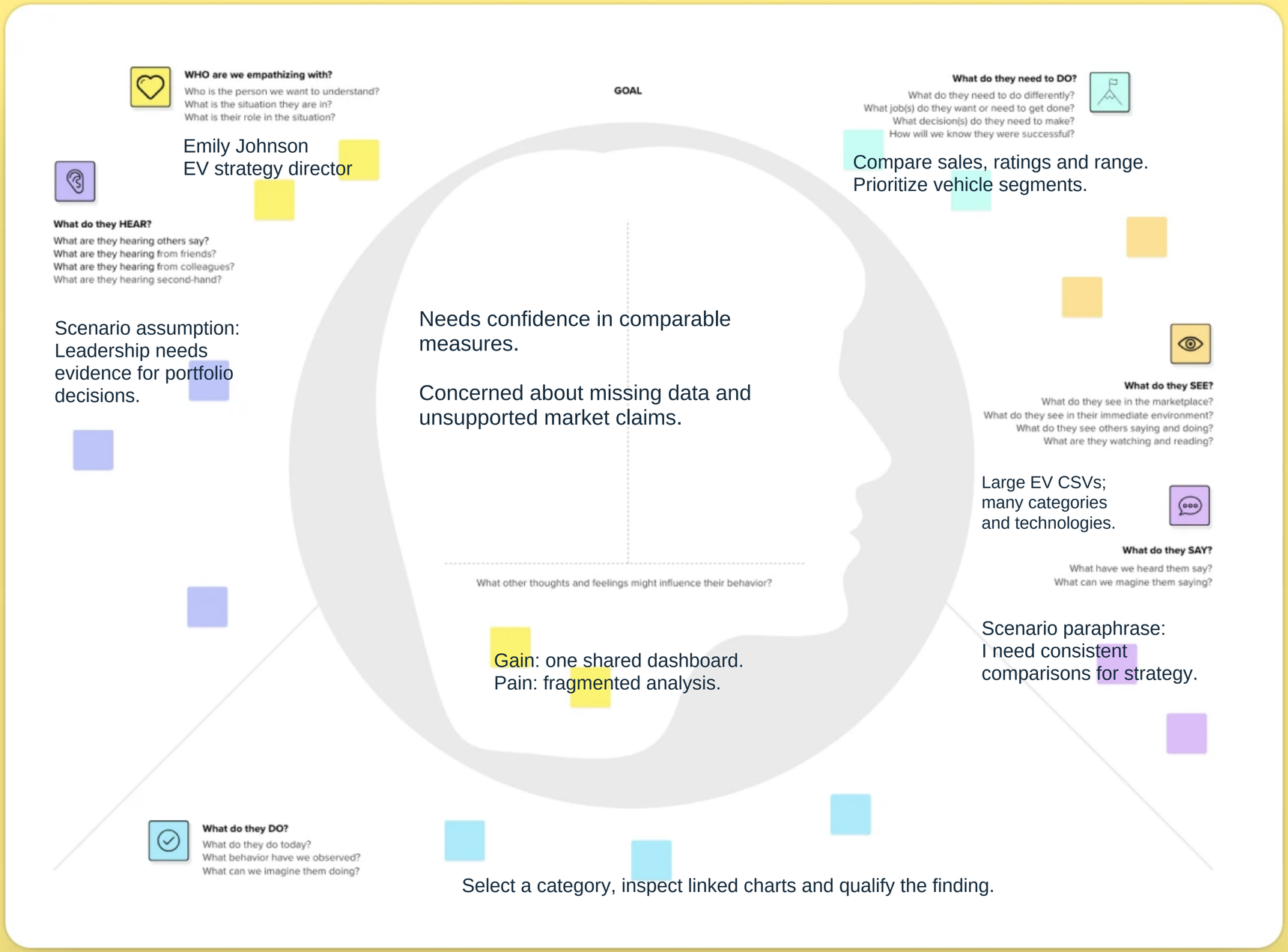
This document records the active participation and roles of each team member during the project demonstration. It ensures that every member contributes meaningfully to the presentation and that responsibilities are distributed fairly and clearly.

| S.No | Team Member Name | Role in Demo | Section Presented | Contribution Summary | Participation (Active / Passive) |
|---|---|---|---|---|---|
| 1 | Vishwas Choyal | Individual author | All sections | Dashboard and documentation | Recorded |
| 2 | Not applicable | Solo project | Not applicable | Not applicable | Not applicable |
| 3 | Not applicable | Solo project | Not applicable | Not applicable | Not applicable |
| 4 | Not applicable | Solo project | Not applicable | Not applicable | Not applicable |
| 5 | Not applicable | Solo project | Not applicable | Not applicable | Not applicable |
| 6 | Not applicable | Solo project | Not applicable | Not applicable | Not applicable |

### Team Coordination Notes:

| Aspect | Response |
|---|---|
| Team Leader / Coordinator | Vishwas Choyal |
| Overall Team Coordination Rating (1-5) | Not applicable to an individual project. |
| Any issues during demo | No recording failure observed. |
| How issues were resolved | Not applicable; the captured video was verified. |

# Emily Johnson empathy canvas




Scenario-derived assumptions, not interviews. Source template: Mural empathy map canvas.

# EV project problem solution fit canvas

## 1. CUSTOMER SEGMENT(S)

EV strategy directors, BI managers and mobility researchers.

## 6. CUSTOMER STATE LIMITATIONS

Time, missing data and synthetic scope.

## 5. AVAILABLE SOLUTIONS

Raw CSVs and static spreadsheets; repeated manual comparisons.

## 2. PROBLEMS / PAINS + frequency

Compare demand, competitors, technology and ratings.
Recurring need at analytical reviews.

## 9. ROOT / CAUSE of every problem

Scattered raw data and inconsistent definitions.
No linked view across the questions.

## 7. BEHAVIOR + its intensity

Compare categories, inspect tooltips and reconcile totals before conclusions.

## 3. TRIGGERS TO ACT

Portfolio reviews and research questions.

## 10. YOUR SOLUTION

Nine Tableau views, category filtering, scenario story and documented validation.

© Daria Nepriakhina / IdeaHackers.nl

## 8. CHANNELS OF BEHAVIOR online + offline

Online: Tableau Public and GitHub.
Offline: TWBX and PDF reports.

## 4. EMOTIONS before & after

Uncertain before; clearer comparisons after.

Scenario-derived responses. Source: IdeaHackers canvas by Daria Nepriakhina; original attribution retained.